



Nova-Band: An AIoT-Driven Upper-Arm Wearable for Real-Time Physiological Monitoring and Running Performance

Maheswara Nala Wisadewa, Almira Edgina Nareswari Haryadi,
Azra Daniswara Tyandaru, Dela Astiara, Rizkyta Aufa
Mutiagassania, Satriya Javas Wibisono

Collaborate with:



Meet The Problem



This lack of physical activity poses a serious threat because it is one of the leading risk factors for premature death worldwide. If left unchecked, this sedentary behavior will lead to a fitness and productivity decline.

The crisis of sedentary lifestyles in society is becoming increasingly evident

37,4%

Indonesians aged 10 and older failing to meet physical activity recommendations.



Worried well anxiety



Physical harm



Overtraining risk



Data inaccuracy

Trends in Runner Injuries: The growing popularity of running among urban populations is often accompanied by lifestyle pressures, the commodification of capitalism, and Fear of Missing Out (FoMO) behavior. The tendency for runners to push themselves beyond their physical limits (overtraining) has led to a shift in the nature of injuries toward more severe soft-tissue and tendon injuries—such as Achilles tendinopathy, calf strains, hamstring and quadriceps injuries, iliotibial band syndrome (ITBS), and knee meniscus tears.

Problem Formulation & Research Objective

Problem Formulation

- How can NovaBand be designed and tested to accurately extract a Readiness Score for preventing soft-tissue injury and cardiovascular collapse in runners?
- How can motion artifacts caused by arm movement affect the accuracy of heart rate and motion data in wrist-based wearable devices used by runners?

Research Objective

- To design and develop NovaBand as an upper-arm wearable that mechanically stabilizes sensor placement to minimize motion artifacts and produce accurate PPG and IMU data.
- To evaluate NovaBand's effectiveness through MAPE accuracy testing and SUS usability testing in extracting a reliable Readiness Score to prevent injury and cardiovascular collapse.

State Of The Arts

Siradj (2016)

Wearables help runners track physical activity practically and independently for self-motivation.



Consumer device accuracy remains low MAPE ranging from 1–42% in step counting and 15–44% in calorie estimation.

Al-Marroof et al (2024)

Smartwatches enable continuous tracking of physiological metrics and support self-monitoring interventions.



Reliability and validity across devices remain inconsistent, especially during high-intensity activity.

Moghaddam et al (2025)

Anatomical placement studies show wrist-worn devices are practical for everyday heart rate monitoring.



Wrist-based PPG sensors are highly vulnerable to motion artifacts, causing accuracy to drop sharply during high-intensity running.

McGrath & Doyle et al (2023)

IMU sensors combined with machine learning can estimate ground reaction force and running kinematics with high precision.



These IMU research systems are largely lab-grade/non-wearable and not yet integrated for real-time daily use by runners.

- Upper-arm placement reduces motion artifacts
- Integrated PPG, IMU, and ESP32 for combined cardio and biomechanical monitoring
- IoT dashboard alerts to organizers early to prevent runners from overtraining risk

The Solution

AI & IoT Synergy

A smart ecosystem that integrates physical sensor readings (IoT) from the device with artificial intelligence (AI) to continuously process physiological signals.



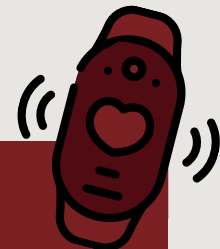
Shared Safety Net

A dual-layer protection system that safeguards users by sending alerts simultaneously to both the individual and the supervisor to prevent collapse.



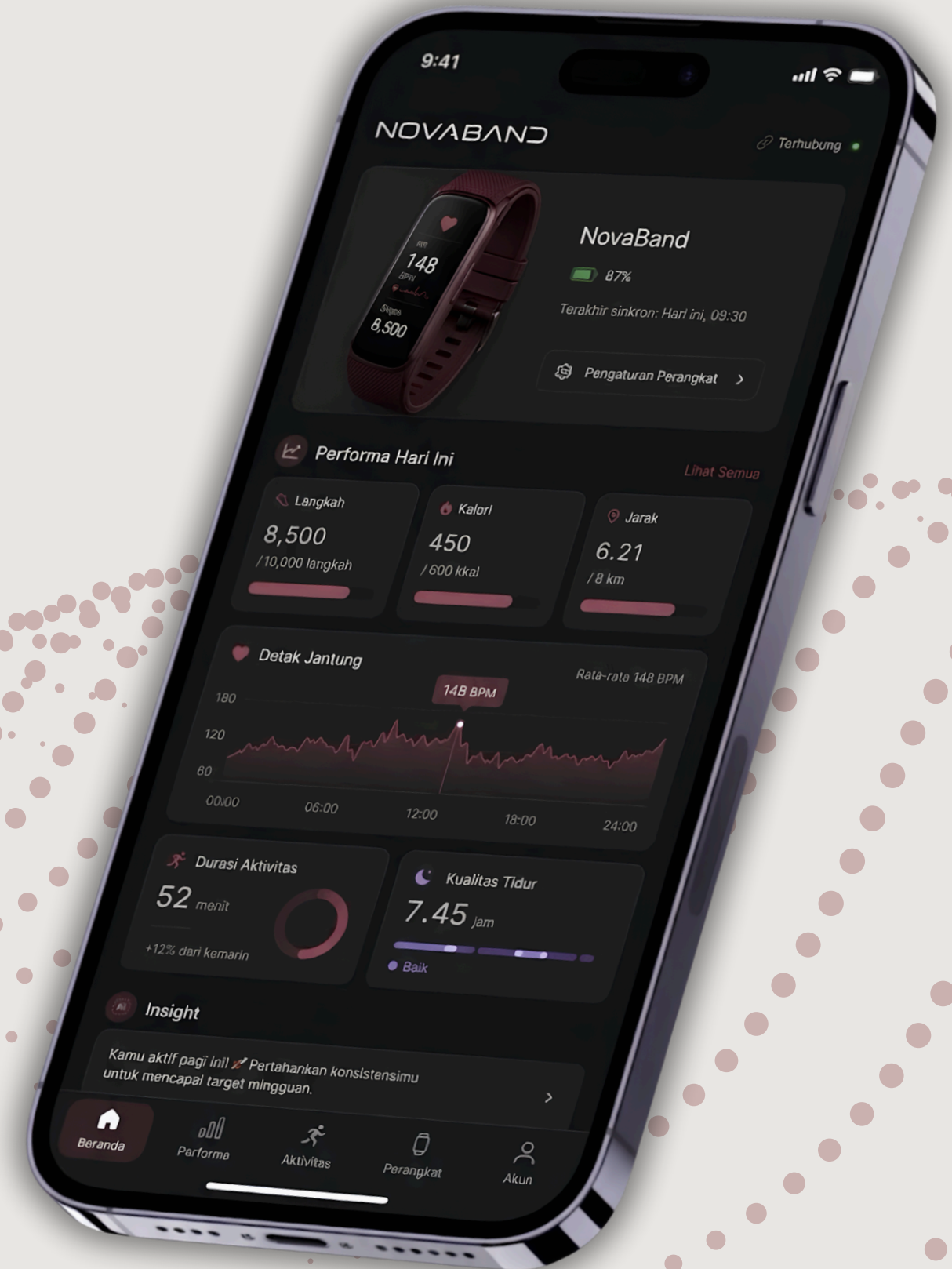
Biological Early Warning

A smart ecosystem that integrates physical sensor readings (IoT) from the device with artificial intelligence (AI) to continuously process physiological signals.



Soft Network Load

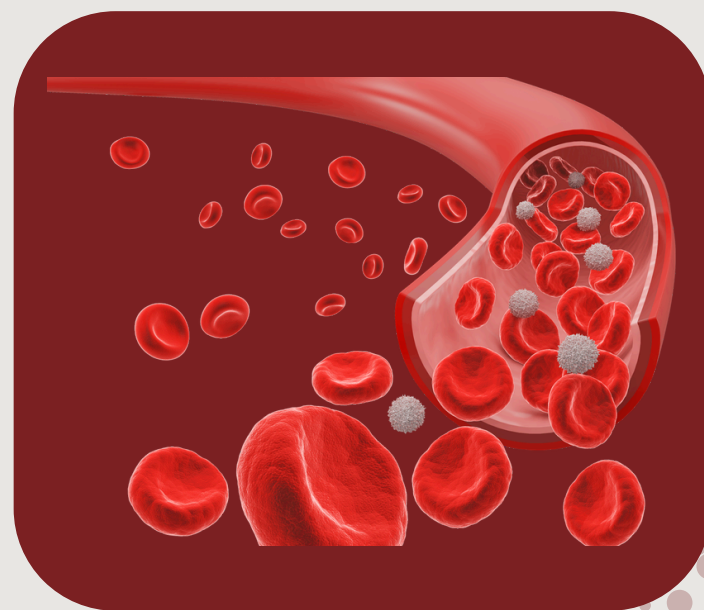
6-axis IMU kinematic tracking measuring mechanical stress on muscles and tendons to prevent Achilles tendinopathy & ITBS.



Hardware Operation (PPG)



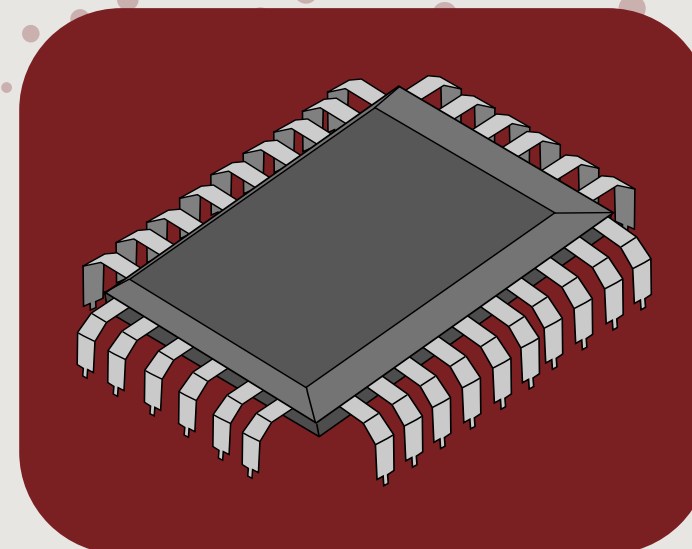
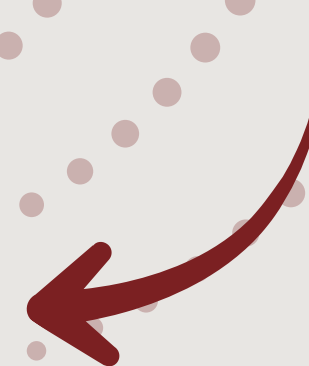
LED Sensor



Blood Absorbs Light



Photodetector

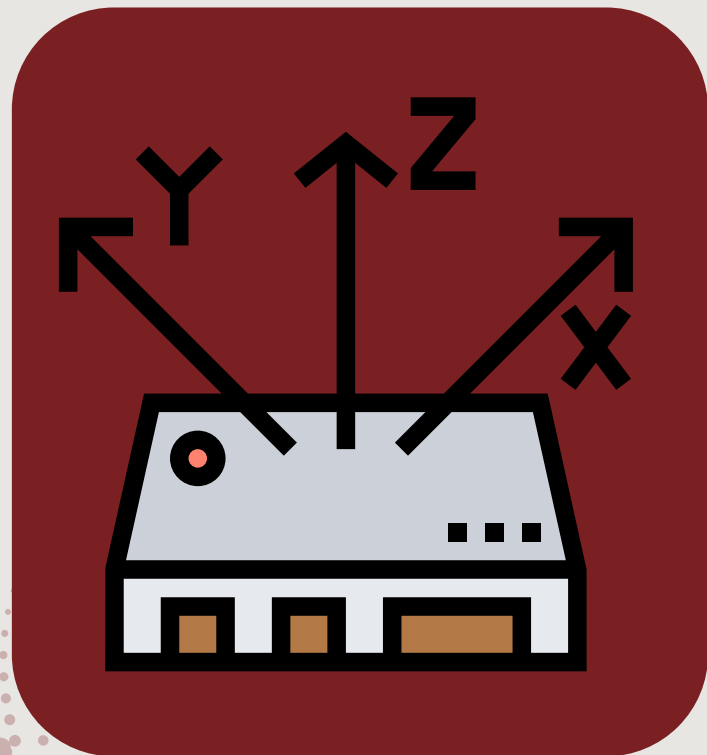


Mikrokontroler



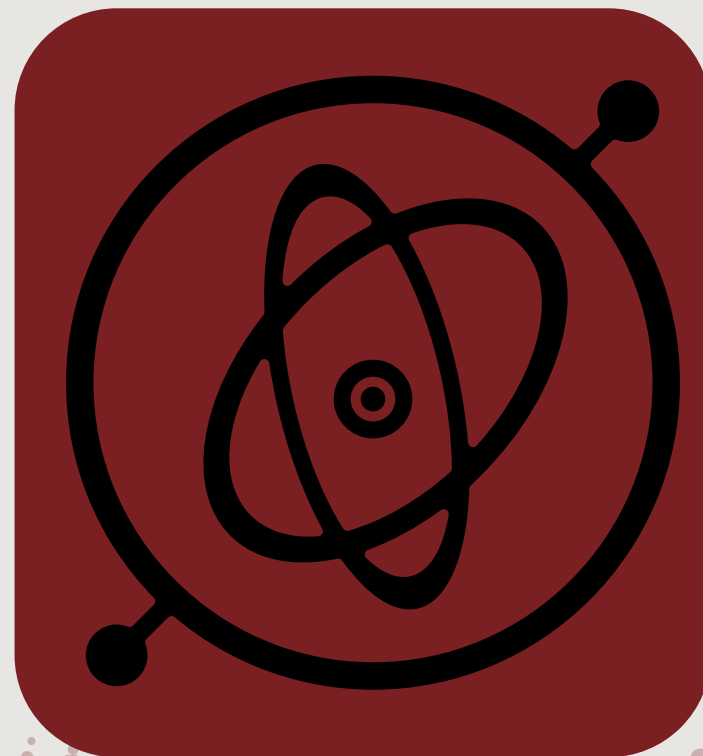
Heart Beats

Hardware Operation (IMU)



Accelerometer

Detects drastic changes in motion speed to identify sudden falls or physical collapse



Gyroscope

Monitors body posture and tilt to detect loss of balance, dizziness, or unsteady movements



Magnetometer

Determines precise body orientation to assist in spatial tracking during emergency situations

Software (IoT Dashboard)



Project Timeline

1. Design Stage

2. Integration Stage

3. Validation Stage

Process Documentation:



Partnership ISIF 2026

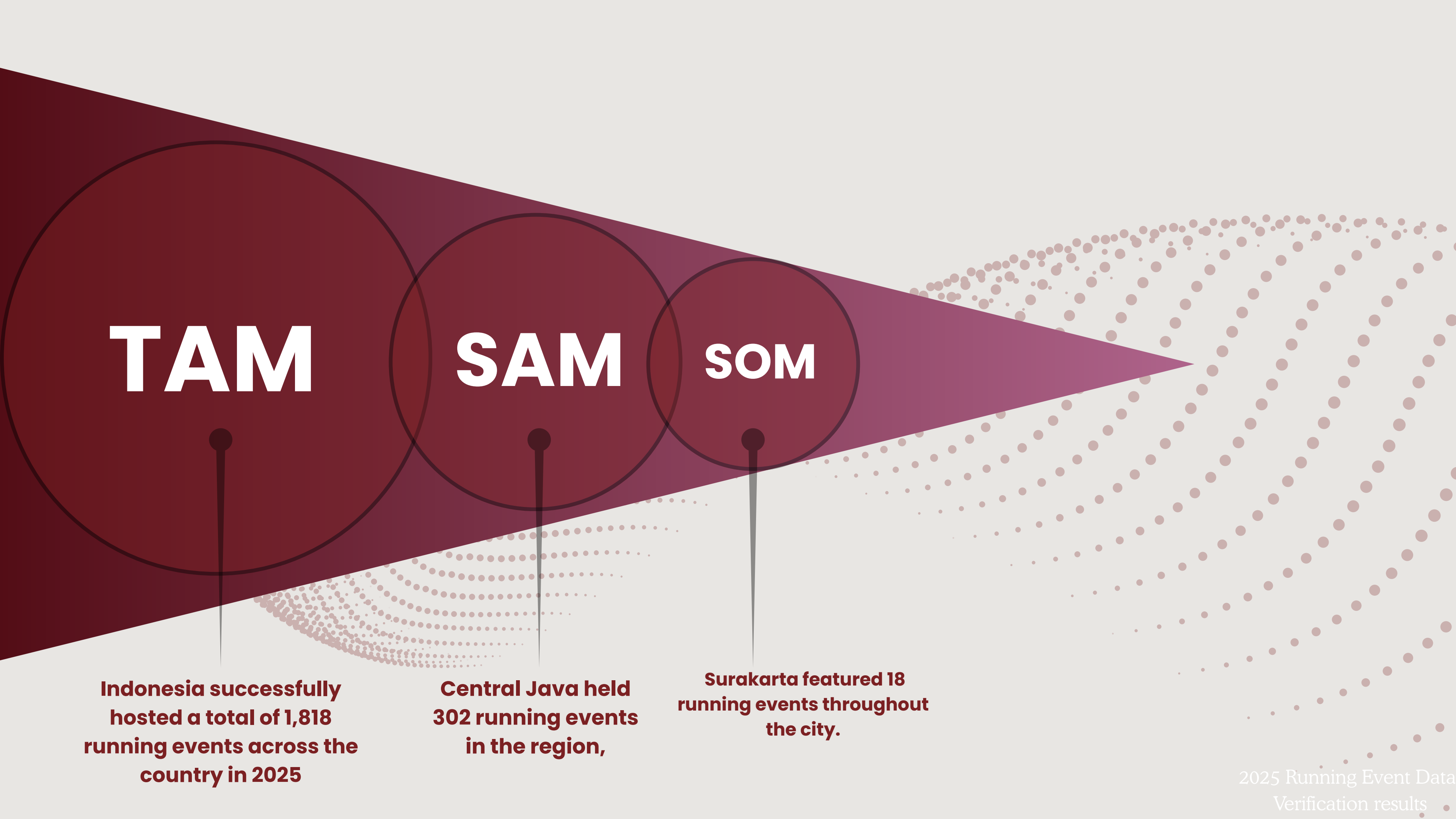


SMAN 1 Surakarta



SMAN 5 Yogyakarta



The infographic features a large, dark red, wedge-shaped graphic on the left side, tapering towards the right. Inside this wedge are three overlapping circles of the same color, labeled TAM, SAM, and SOM from left to right. Below each circle is a vertical line with a dot at the top, pointing to a text box. The background is a light gray with a pattern of small, dark gray dots that form a large, stylized 'S' shape, suggesting a running path or data flow. The text is in a clean, sans-serif font, with the region names in large, bold, white letters and the descriptions in smaller, dark red letters.

TAM

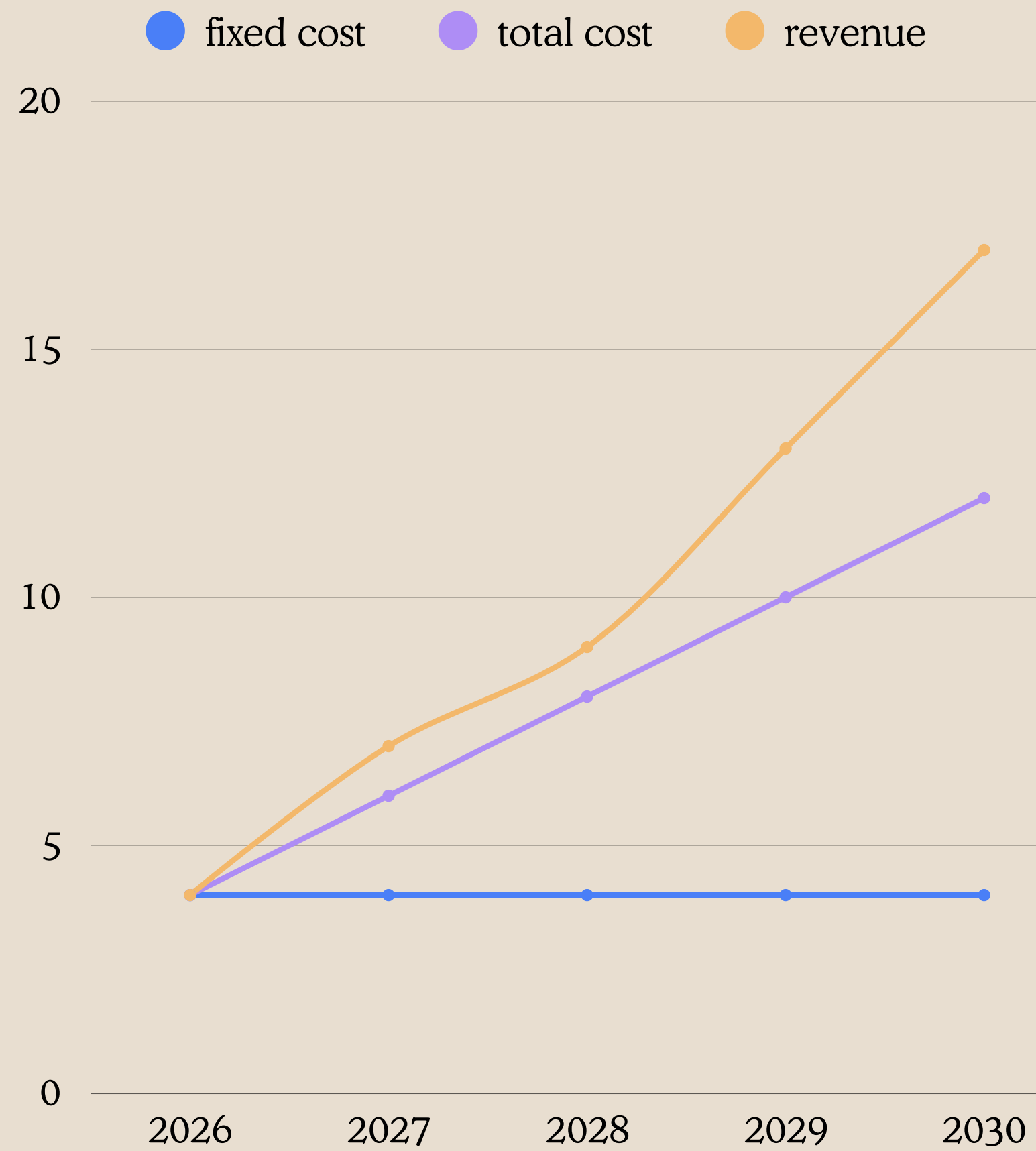
**Indonesia successfully
hosted a total of 1,818
running events across the
country in 2025**

SAM

**Central Java held
302 running events
in the region,**

SOM

**Surakarta featured 18
running events throughout
the city.**



Implementation



Result & Discussion

Hardware Accuracy

Based on test results, the NovaBand sensor readings show a MAPE value ranging from **2.5% to 3.5%**. MAPE measures the average magnitude of sensor reading errors; thus, a lower value indicates greater accuracy. These results demonstrate that the NovaBand sensor has a low error rate and provides stable readings, even during high-intensity activities.



Usability Evaluation (SUS)

The obtained SUS score is **84.5** out of 100. This score indicates that NovaBand's usability is excellent and acceptable to users. Based on SUS interpretation, the score falls into the "Acceptable" category with a Grade A, meaning NovaBand is considered easy to use and comfortable for users.

84.5%

Conclusion

System Integration

NovaBand successfully integrates PPG (heart rate) and IMU (body movement) sensors powered by an ESP32 microcontroller positioned on the upper arm to collect physiological and motion data simultaneously.

AI Analysis

Leverages AI to process activity patterns and physiological responses together in real time during training.

Impact and Final Result

NovaBand serves as an accurate, practical, ergonomic, and data-driven wearable innovation for real-time runner performance monitoring.

Project – 2

2025

Interior Design

Minimalist Bedroom Design

Designed a cozy minimalist bedroom with soft color palettes, efficient storage, and modern interior elements. The project focused on creating a relaxing and functional space that combines comfort, simplicity, and modern aesthetics to support daily living and better organization.

